The Brain as a Least-Action-Like Cognitive System: Cognitive Path Optimization Under Finite Resources

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Abstract

This paper proposes a least-action-like model of cognition. The brain is not treated as an unlimited computational system searching for absolute truth, but as a finite-resource adaptive system selecting cognitive paths that reduce future uncertainty, support action, and preserve adaptive self-environment coupling under constraints of energy, attention, time, and memory capacity. The claim is not that the biological brain literally obeys the physical principle of least action. Rather, the paper develops a resource-constrained active inference framework. In this framework, neural integration supplies a microscopic dynamical basis, variational free energy supplies an instantaneous cost function, resource constraints generate a path-level cost functional across time, multi-model Bayesian competition explains the dynamic selection among candidate interpretations, memory dynamics preserve reusable low-cost paths, and the higher-order observer is formalized as a meta-control system that modulates cost weights, precision, update gains, and effective temperature.

Keywords: active inference; free energy principle; least action; cognitive action; memory dynamics; resource constraints; predictive processing; meta-control; abstraction; cognitive overload

1 Starting Point

In classical physics, the principle of least action states that the actual trajectory of a system is not arbitrary. Under suitable constraints, the system follows a path for which an action functional is stationary or optimal.

An analogous, but not identical, idea can be introduced for cognition:

> The brain is not an unlimited computational system. It is a finite-resource system that selects cognitive paths that are sufficiently stable, sufficiently low-cost, able to explain the world, and useful for action.

Here "action" is not the physical integral of kinetic minus potential energy. It is the accumulated cost paid by a cognitive system across time. This cost includes prediction error, neural energy consumption, attentional load, memory maintenance cost, higher-order control cost, and action cost.

In this sense, the brain may be understood as a least-action-like cognitive system. More precisely:

> Intelligence is not unlimited computation, but path selection under finite resources.

2 Relation to Existing Work

The present framework is not intended to replace existing theories of bounded cognition, cognitive control, compression, or active inference. Its purpose is to connect them through a single path-cost formulation.

First, the finite-resource assumption is closely related to resource-rational analysis, especially the work of Lieder and Griffiths. Resource-rational analysis extends Bayesian and rational models of cognition by asking which cognitive operations are available to an agent and how costly those operations are in time and computational resources. The present model inherits this spirit, but embeds resource costs inside an active-inference path functional rather than treating them only as constraints on isolated computations.

Second, the treatment of higher-order control is closely aligned with the Expected Value of Control theory developed by Shenhav, Botvinick, and Cohen. EVC proposes that control allocation depends on expected payoff, the amount of control required, and the cognitive effort cost. In the present framework, the term $\lambda_C C_t$ and the meta-control variable θ_t generalize this idea into a temporally extended path-selection model.

Third, the claim that abstraction is a compressed reusable structure resonates with minimum description length and compression-based theories of learning and curiosity. Rissanen's MDL principle treats good models as those that shorten the description of data. Schmidhuber's theory of compression progress treats improvements in compressibility as a driver of curiosity and discovery. The present framework adds a memory-dynamical interpretation: abstractions persist when they reduce future free energy across multiple contexts by more than their maintenance cost.

Finally, the free energy principle and active inference provide the formal background for interpreting perception, action, attention, and learning as forms of uncertainty reduction. The contribution of this paper is to combine active inference with resource-rational costs, control valuation, memory retention, abstraction, and overload into a single least-action-like cognitive path model.

3 From Free Energy to Cognitive Action

In the free energy principle and active inference, a system minimizes variational free energy in order to make its internal beliefs approximate the generative structure of the external world.

Let the hidden environmental state be

$$x_t \in \mathcal{X}, \tag{1}$$

and the observation be

$$o_t \in \mathcal{O}. \tag{2}$$

The system maintains an approximate posterior

$$q(x_t). \tag{3}$$

For a generative model m^k, the instantaneous variational free energy can be written as

$$F_t^k[q^k] = \mathbb{E}_{q^k}\left[\ln q^k(x_t) - \ln p(o_t, x_t|o_{1:t-1}, m^k)\right]. \tag{4}$$

It can also be decomposed as

$$F_t^k[q^k] = D_{KL}\left[q^k(x_t)\|p(x_t|o_{1:t}, m^k)\right] - \ln p(o_t|o_{1:t-1}, m^k). \tag{5}$$

Since the KL divergence is non-negative,

$$F_t^k[q^k] \geq -\ln p(o_t|o_{1:t-1}, m^k). \tag{6}$$

Thus, minimizing variational free energy makes the approximate posterior closer to the Bayesian posterior and increases the explanatory adequacy of the model for the current observation.

However, the brain does not minimize error only at an isolated instant. It must maintain perception, memory, action, and control continuously over time. The instantaneous cost must therefore be extended into a path-level cost.

Expected free energy is normally a future-oriented quantity. To avoid mixing an instantaneous present-time free energy term with a future policy evaluation term, define a rolling-horizon expected free energy:

$$G_{t:t+H}(\pi_t) = \int_t^{t+H} G_\tau(\pi_t)d\tau. \tag{7}$$

In the path functional below, let

$$g_t(\pi_t) = \frac{1}{H} G_{t:t+H}(\pi_t) \tag{8}$$

denote the local expected free-energy rate associated with the policy considered at time t. This keeps the integrand dimensionally and temporally consistent.

Define a resource-constrained cognitive action functional:

$$\mathcal{A}_{cog}[q,\pi] = \int_{t_0}^{t_1} \left[F_t(q_t) + \beta g_t(\pi_t) + \lambda_E E_t + \lambda_A A_t + \lambda_M M_t + \lambda_C C_t\right] dt. \tag{9}$$

Here F_t is current variational free energy; $g_t(\pi_t)$ is the local expected free-energy rate induced by a rolling future horizon; E_t is energetic cost; A_t is attentional cost; M_t is memory maintenance and retrieval cost; C_t is higher-order control cost; and $\lambda_E, \lambda_A, \lambda_M, \lambda_C$ are resource cost weights.

Cognitive path selection can then be written as

$$(q^*, \pi^*) = \arg\min_{q,\pi} \mathcal{A}_{cog}[q,\pi]. \tag{10}$$

This is not a direct equivalence to physical action. It is a path-cost functional for a resource-constrained active inference system.

4 The Brain Maintains Multiple Candidate Paths

Cognition does not usually begin with a single final answer. In complex situations, the brain maintains several candidate interpretations:

$$H_1, H_2, \ldots, H_K. \tag{11}$$

Each candidate interpretation corresponds to a generative model m^k and has a belief weight

$$\rho_t^k = p(m^k | o_{1:t}), \tag{12}$$

with

$$\rho_t^k \geq 0, \qquad \sum_{k=1}^{K} \rho_t^k = 1. \tag{13}$$

The overall belief can be represented as a mixture:

$$q(x_t) = \sum_{k=1}^{K} \rho_t^k q^k(x_t). \tag{14}$$

If each path has a total instantaneous cost

$$\ell_t^k = F_t^k + \beta g_t^k + \lambda_E E_t^k + \lambda_A A_t^k + \lambda_M M_t^k + \lambda_C C_t^k, \tag{15}$$

then model weights can be updated by an exponential rule:

$$\rho_{t+\Delta t}^{k}=\frac{\rho_t^k\exp(-\ell_t^k\Delta t)}{\sum_j \rho_t^j\exp(-\ell_t^j\Delta t)}. \tag{16}$$

Taking the continuous-time limit $\Delta t \to 0$ yields the replicator equation:

$$\frac{d\rho_t^k}{dt}=\rho_t^k(\bar{\ell}_t-\ell_t^k), \tag{17}$$

where

$$\bar{\ell}_t=\sum_j \rho_t^j\ell_t^j. \tag{18}$$

This means:

> Interpretive paths whose cost is below the average gain weight, while paths whose cost is above the average lose weight.

This helps explain several cognitive phenomena: hesitation, insight, association, narrowed thinking under fatigue, and reduced abstraction under stress. The cognitive state of the brain is therefore not a single fixed answer. It is a dynamic competition among candidate interpretations under finite resource constraints.

5 Neural Integration as a Microscopic Basis

A single neuron can be described, in a simplified form, as a temporal integrator:

$$\tau\frac{dV}{dt}=-(V-V_{rest})+I(t). \tag{19}$$

When membrane potential reaches threshold,

$$V(t)\geq V_{th}, \tag{20}$$

the neuron fires.

Thus the neuron is already a time-integration and threshold-triggering system:

$$\text{input accumulation}\to\text{threshold crossing}\to\text{spiking}. \tag{21}$$

At the cognitive level, the activation strength b of a belief, memory, or circuit can be described by a coarse-grained phenomenological model:

$$\tau\dot{b}=-\alpha b-\Gamma_b\frac{\partial F}{\partial b}+\gamma V_{value}-\eta E+\xi_t. \tag{22}$$

Here b is the activation strength of a belief or circuit; $-\partial F/\partial b$ is the free-energy gradient acting on that activation; V_{value} is a value signal; E is energetic or resource cost; $\Gamma_b, \alpha, \gamma, \eta$ are regulatory parameters; and ξ_t is unmodeled noise or fluctuation.

Because prediction errors are often the main components of free-energy gradients in predictive processing models, this can be approximated heuristically as

$$\tau \frac{db}{dt} \approx -\alpha b + \beta \varepsilon + \gamma V_{value} - \eta E. \tag{23}$$

This second expression should be read as a phenomenological summary, not as a strict derivation from the free energy principle.

This suggests:

> The nervous system integrates inputs, errors, value, and energetic costs over time in order to determine which circuits are activated, reinforced, inhibited, or forgotten.

Neural firing integration can therefore be understood as a microscopic dynamical basis for free-energy minimization and cognitive path selection.

6 Memory Is Not Recording, but the Preservation of Low-Cost Paths

The brain does not compute everything from scratch. It preserves useful paths from the past so that similar situations can be handled at lower cost in the future.

Memory is therefore not a complete recording of the past. It is a compressed, selective, and reconstructive preservation of past cognitive paths.

Let the strength of a memory trace be

$$m_i(t) \in [0, 1]. \tag{24}$$

Its dynamics can be written as

$$dm_i = [-\lambda_i m_i + \alpha r_i(t) + \beta_m \max(0, \Delta F_i(t))]\, dt + \sigma_m dW_t^i, \tag{25}$$

where

$$\Delta F_i(t) = F_{\text{baseline}}(t) - F_{\text{with } m_i}(t). \tag{26}$$

This is the reduction in free energy produced by the memory trace. If $\Delta F_i(t) > 0$, the memory helps explain the current situation. If $\Delta F_i(t) \leq 0$, it provides no explanatory benefit and may even interfere.

The term $\max(0, \Delta F_i(t))$ is a local, instantaneous consolidation signal. The longer-term retention rule can be understood as the value function generated by such local rewards. Define an instantaneous mnemonic reward as

$$r_i^{mem}(t) = \max(0, \Delta F_i(t)) - c_i(m_i, t), \tag{27}$$

where $c_i(m_i, t)$ is the ongoing maintenance cost of the trace. Then the long-term value of retaining the memory is

$$V_i(t) = \mathbb{E}\left[\int_t^\infty e^{-\gamma(s-t)} r_i^{mem}(s) ds\right]. \tag{28}$$

The retention rule for a memory can be written as

$$\text{keep } m_i \iff V_i(t) > 0, \tag{29}$$

or equivalently,

$$\mathbb{E}\left[\int_t^\infty e^{-\gamma(s-t)} \Delta F_i(s) ds\right] > C_{\text{maintain}}(m_i). \tag{30}$$

Thus:

> What is preserved in the long run is not the fullest detail, but the structure that repeatedly reduces future free energy at a maintenance cost lower than its benefit.

This also explains why abstraction emerges. Abstraction is not an additional faculty imposed on memory. It is a natural result of memory dynamics under finite resources. When several experiences share a structure, and that structure repeatedly reduces free energy across situations, it becomes more likely to be preserved.

Define the value of an abstract structure a as

$$V(a) = \sum_j w_j \Delta F_a(e_j) - C_{\text{maintain}}(a). \tag{31}$$

If

$$V(a) > 0, \tag{32}$$

then the abstract structure is worth preserving.

Therefore:

> An abstraction is a compressed structure that reduces free energy across multiple experiences and whose maintenance cost is lower than its cumulative benefit.

7 The Brain Does Not Minimize Immediate Effort Alone

A common misunderstanding is that the brain simply chooses the easiest or least effortful path. This is only partly correct. A more precise claim is:

> The brain tends to form circuits that have lower accumulated free energy over the long run. These circuits are not necessarily locally simplest, but they must stabilize a tradeoff among prediction error, energy consumption, memory maintenance cost, and action value.

Many advanced abilities are costly at the beginning: learning mathematics, programming, language, or abstract modeling is initially demanding. Once a stable circuit is formed, however, future processing of similar problems becomes substantially cheaper.

The brain therefore does not simply select

$$\text{the currently easiest path.} \tag{33}$$

It selects

$$\text{the path with lower accumulated long-term cost.} \tag{34}$$

Formally:

$$C^* = \arg\min_C \int_{t_0}^{t_1} \left[F_C(t) + \lambda_E E_C(t) + \lambda_M M_C(t) + \lambda_C C_C(t)\right] dt. \tag{35}$$

8 Overload, Fatigue, and Attentional Collapse

Let the current total cost rate be

$$L(t) = F_t + \beta g_t + \lambda_E E_t + \lambda_A A_t + \lambda_M M_t + \lambda_C C_t. \tag{36}$$

Let the current resource boundary be

$$R_{\text{total}}(t). \tag{37}$$

When

$$L(t) < R_{\text{total}}(t), \tag{38}$$

the system can maintain stable inference, memory integration, and action control.

When

$$L(t) \geq R_{\text{total}}(t), \tag{39}$$

the system enters overload.

The brain may then adopt protective strategies: lowering input gain, narrowing attention, falling back to familiar circuits, reducing abstract reasoning, reducing higher-order integration, or producing fatigue, brain fog, dizziness, and avoidance.

An effective temperature can be introduced:

$$T_{\text{eff}}(t) = T_0 \max\left(1, \frac{L(t)}{R_{\text{total}}(t)}\right). \tag{40}$$

As load approaches or exceeds the resource boundary, noise increases and belief updating becomes less stable:

$$d\mu_t^k = -\Gamma_\mu \nabla_\mu F_t^k dt + \sqrt{2\Gamma_\mu T_{\text{eff}}(t)} dW_t^{\mu,k}. \tag{41}$$

This provides a formal model of cognitive overload.

However, caution is required. Dizziness, fatigue, and brain fog in real organisms can involve vestibular function, blood pressure, autonomic regulation, metabolic state, inflammation, sleep, and other physiological factors. The present model should not be treated as a unique causal account of these phenomena. It is a testable cognitive-level explanatory hypothesis.

9 The Higher-Order Observer as Meta-Control

In this framework, the higher-order observer should not be interpreted as a little person inside the brain watching everything.

More rigorously, it is a set of slow meta-control variables that regulate lower-level inference.

Define the meta-control state as

$$\theta_t = \{\lambda_E, \lambda_A, \lambda_M, \lambda_C, \Gamma_\mu, T_{\text{eff}}, \kappa\}. \tag{42}$$

These variables regulate resource cost weights, belief update speed, attentional precision, exploration noise, the competition temperature among candidate paths, and higher-order control gain.

The meta-control objective can be written as

$$J(\theta) = \mathbb{E}\left[\int_0^T \ell_t(\theta) dt\right]. \tag{43}$$

A meta-learning dynamics can be written as

$$d\theta_t = -\eta_\theta \nabla_\theta J(\theta_t) dt + \sigma_\theta dW_t^\theta. \tag{44}$$

Thus, the rigorous version of the higher-order observer is:

> A meta-control system that modulates precision, cost weights, update gains, and effective temperature.

Its role is not to mysteriously "see consciousness", but to redistribute finite resources among candidate cognitive paths so that the system tends to remain on lower long-term cost trajectories.

10 Unified Model

The preceding elements can be summarized as a resource-constrained active inference system:

$$\begin{aligned}
d\mu_t^k &= -\Gamma_\mu \nabla_\mu F_t^k dt + \sqrt{2\Gamma_\mu T_{\text{eff}}(t)}dW_t^{\mu,k} \\
dz_t^k &= -\ell_t^k dt + \sigma_\rho dW_t^{z,k} \\
\rho_t^k &= \frac{e^{z_t^k}}{\sum_j e^{z_t^j}} \\
dm_i &= [-\lambda_i m_i + \alpha r_i(t) + \beta_m \max(0, \Delta F_i(t))]\, dt + \sigma_m dW_t^{m,i} \\
d\theta_t &= -\eta_\theta \nabla_\theta J(\theta_t)dt + \sigma_\theta dW_t^\theta .
\end{aligned} \tag{45}$$

where

$$\ell_t^k = F_t^k + \beta g_t^k + \lambda_E E_t^k + \lambda_A A_t^k + \lambda_M M_t^k + \lambda_C C_t^k . \tag{46}$$

The model integrates the following chain:

$$\begin{aligned}
&\text{finite resources} \to \text{neural integration} \to \text{prediction error} \to \text{free-energy minimization} \\
&\to \text{cognitive action} \to \text{multi-path competition} \to \text{memory dynamics} \to \text{abstraction} \\
&\to \text{meta-control} \to \text{adaptive intelligence.}
\end{aligned} \tag{47}$$

11 Potential Empirical Predictions and Operationalization

The model becomes scientifically useful only if its latent variables can be connected to observable quantities. The cost weights $\lambda_A, \lambda_C, \lambda_M$ and the effective temperature T_{eff} should not be treated as directly observed neural quantities. They are latent variables to be estimated by fitting behavioral and physiological data.

Several empirical predictions follow.

First, under dual-task load, time pressure, sensory noise, or sleep deprivation, the model predicts an increase in effective temperature. Candidate-path selection should therefore become more stochastic and higher entropy:

$$P(k) \propto \exp\left(-\frac{\mathcal{A}_k}{T_{\text{eff}}}\right). \tag{48}$$

As T_{eff} increases, choice distributions should flatten, reaction times should become more variable, and reliance on familiar low-cost strategies should increase.

Second, when control cost increases, the system should allocate less control to high-effort strategies even when those strategies are more accurate. This predicts larger task-switching costs, higher Stroop-like interference, reduced working-memory maintenance, and more habitual responding. Candidate empirical proxies include response time, error rate, subjective effort ratings, pupil diameter, frontal midline theta, and dACC/PFC BOLD activity.

Third, the memory-value rule predicts that traces that reduce future free energy should be preferentially retained and reactivated. This can be tested with delayed recall, recognition, sleep-dependent consolidation, representational reinstatement, and transfer tasks. A trace that improves later prediction or action should show greater retention than an equally vivid trace that does not reduce future uncertainty.

Fourth, the abstraction rule predicts that shared structure across tasks should promote compressed representations. Under resource stress, abstraction depth should decrease: participants should rely more on concrete familiar exemplars and less on higher-level rules. In computational models, this should appear as a shift from higher-level latent variables toward lower-level cached policies.

These predictions do not prove that the biological brain exactly minimizes $\mathcal{A}_{cog}$. They make the framework falsifiable: if manipulations of load, control cost, or memory value do not change choice entropy, control allocation, retention, or abstraction in the predicted directions, the model must be revised.

12 What Can Be Proved and What Remains Hypothetical?

Some parts of the framework are standard or direct mathematical results:

Claim	Status
Variational free energy upper-bounds negative log evidence	Provable
Minimizing free energy approximates Bayesian posterior inference	Provable
Resource constraints yield a Lagrangian path functional	Provable
Softmax weights arise from accumulated cost	Provable
The continuous-time limit yields replicator dynamics	Provable
Logits guarantee legal probability weights	Provable
Gradient flow monotonically decreases free energy	Provable under specified conditions
Memory retention follows a threshold rule	Provable under net-benefit maximization
Abstract structure retention generalizes the memory rule	Derivable from the memory rule

Other parts remain hypotheses:

Claim	Status
The biological brain exactly minimizes this cognitive action	Hypothesis
The specific forms of energetic, attentional, mnemonic, and control costs are unique	Hypothesis
Higher-order meta-control maps cleanly onto a specific brain region	Requires empirical support
Effective temperature precisely represents cognitive overload	Effective model
Dizziness is directly caused by cognitive load exceeding a resource boundary	Not established
Biological memory computes future free-energy benefit exactly	Likely only approximate
Abstraction is always equivalent to low-free-energy compression	Strong theoretical hypothesis

The framework should therefore not be read as a proof that the brain literally works this way. It is better understood as:

> A computable, simulable, and prediction-generating normative model of cognition.

13 Final Thesis

The whole theory can be condensed into one statement:

> The brain does not search for truth through unlimited computation. It searches, under finite resources, for low-action cognitive paths. Neural integration supplies the

> microscopic dynamics, free-energy minimization supplies the error-correction principle, memory dynamics preserve reusable low-cost circuits, and the higher-order observer, understood as a meta-control system, reallocates resources among candidate paths. When cognitive load exceeds the system boundary, low-action paths become difficult to maintain, and the system may show fatigue, attentional collapse, failure of higher-order integration, and cognitive instability.

In a more academic form:

> The brain may be understood as a least-action-like cognitive system. Under finite energetic, attentional, temporal, and mnemonic resources, neural dynamics tend to select perceptual, mnemonic, and action trajectories that minimize accumulated variational free energy while preserving adaptive self-environment coupling.

This is the core of the least-action-like cognitive model.

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